FIMsim Test Cases

Two ready-to-use study areas so you can try FIMsim end to end without any input data of your own. Each test case walks through one complete flood-model setup:

Test case	AOI shapefile	Location	Model	Flood event
1	`AOI_1_Neuse/AOI_1.shp`	Neuse River, North Carolina	**LISFLOOD-FP**	Hurricane Matthew — October 2016
2	`AOI_2_Texas/AOI_2.shp`	Village Creek, Texas	**TRITON**	Hurricane Harvey — August 2017

What you need: FIMsim (web app or desktop installer — see the main README on GitHub) and an internet connection. You provide the AOI shapefile plus the start and end dates of the flood event you want to simulate — all terrain, land-cover, river-network, and discharge data are downloaded automatically.

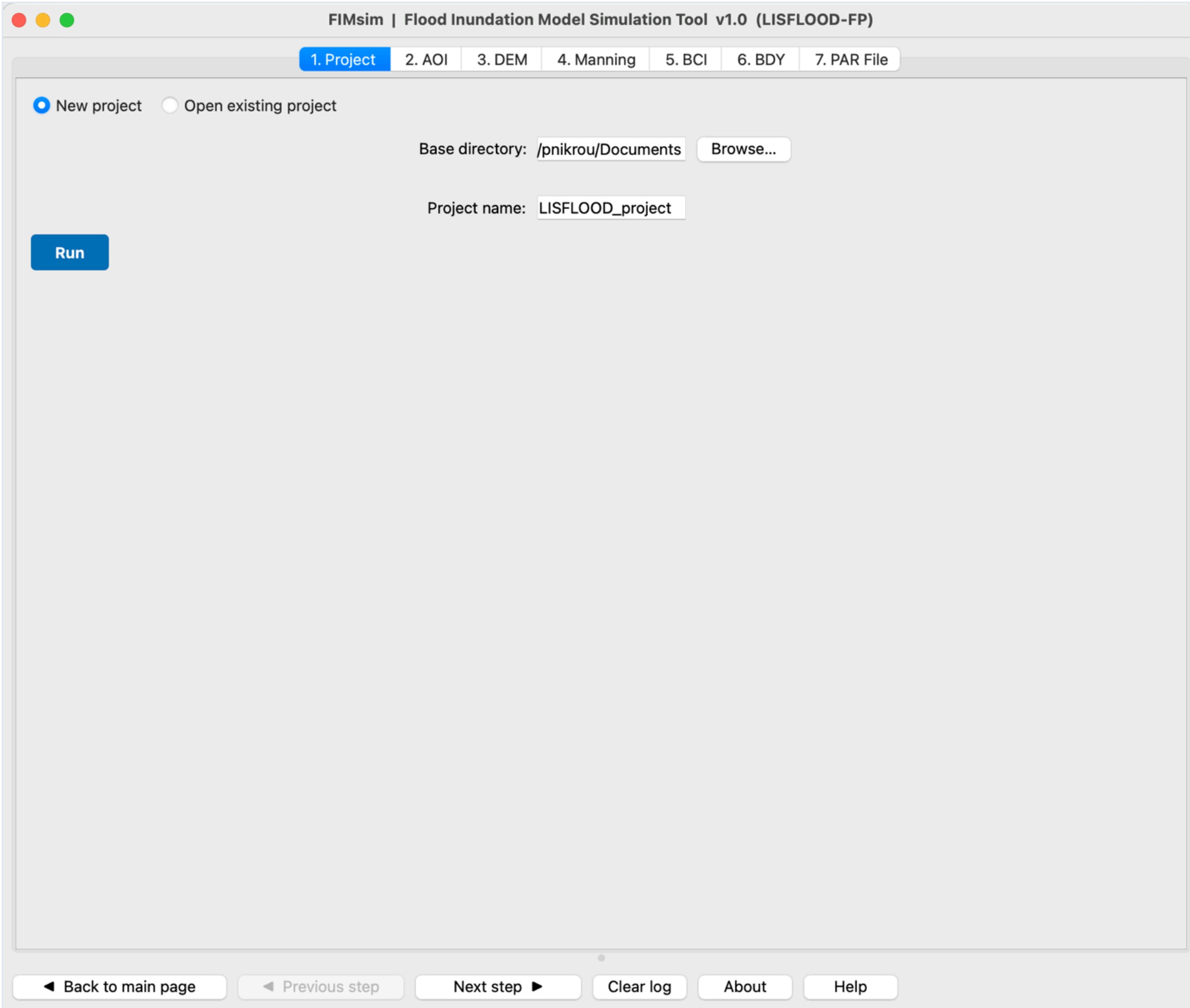
Test Case 1 — LISFLOOD-FP · Neuse River, North Carolina

AOI: `AOI_1_Neuse/AOI_1.shp` · 293.4 km² · CRS EPSG:26917 (NAD83 / UTM 17N) · Event: Hurricane Matthew, 2016-10-05 → 2016-10-20

In early October 2016, Hurricane Matthew brought extreme rainfall to eastern North Carolina, producing record flooding along the Neuse River. This walkthrough builds the complete LISFLOOD-FP input package for a 15-day simulation of that event, using the USGS stream gage inside the AOI for the inflow hydrograph.

Step 1 — Project

Open FIMsim and choose the **LISFLOOD-FP** model. Create a new project — pick any project name and an empty folder where all outputs will be written.

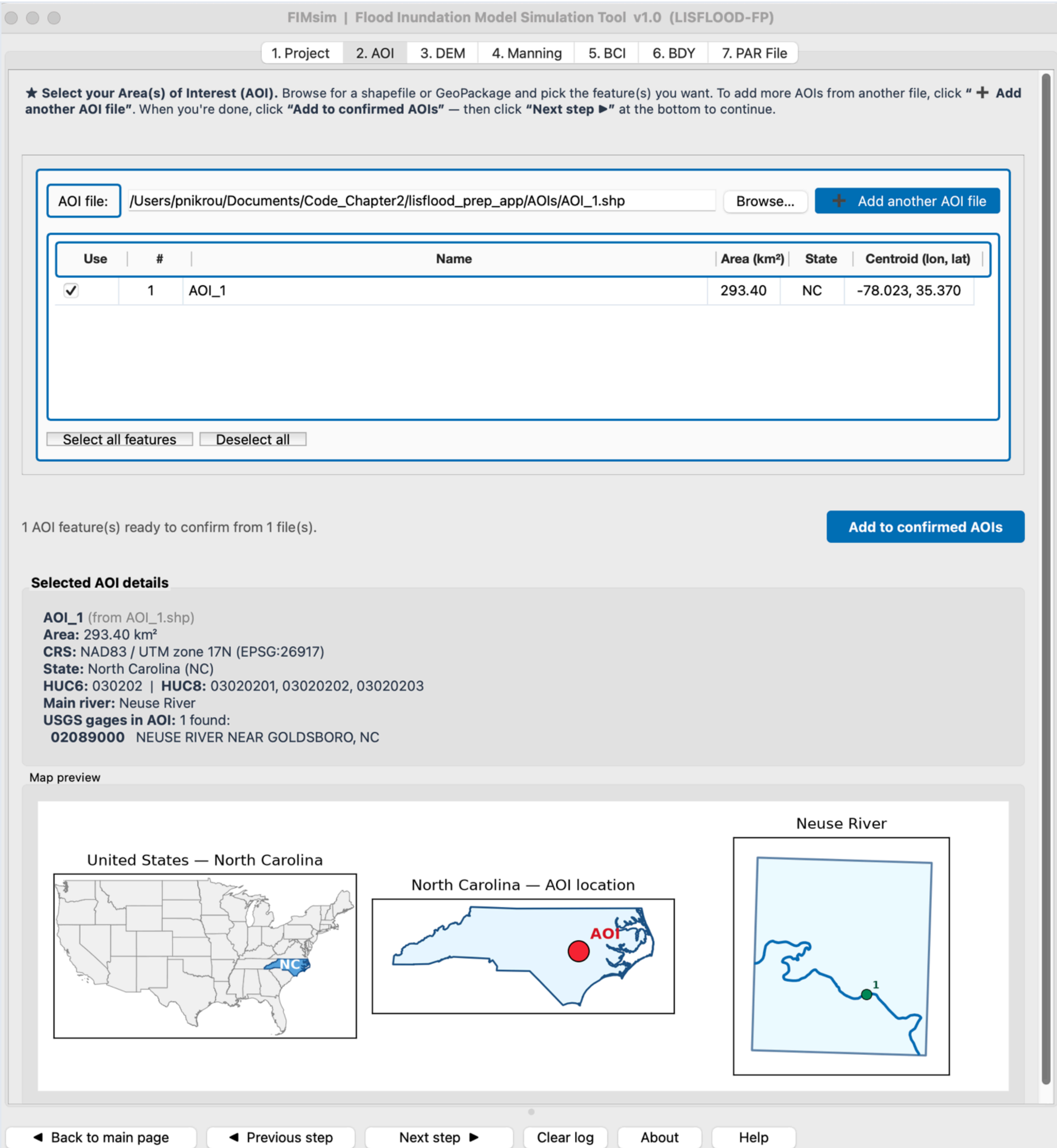


Step 2 — AOI

Click **Browse** and select `AOI_1.shp`. You can load several AOI files, or pick one or more features from a shapefile that contains multiple polygons — for this test case it is a single feature: tick it and click **Add to confirmed AOIs**.

The panel gives an overview of the selected AOI: geographic location, area, HUC6/HUC8 codes, CRS, the main river, and any USGS gages found inside the AOI.

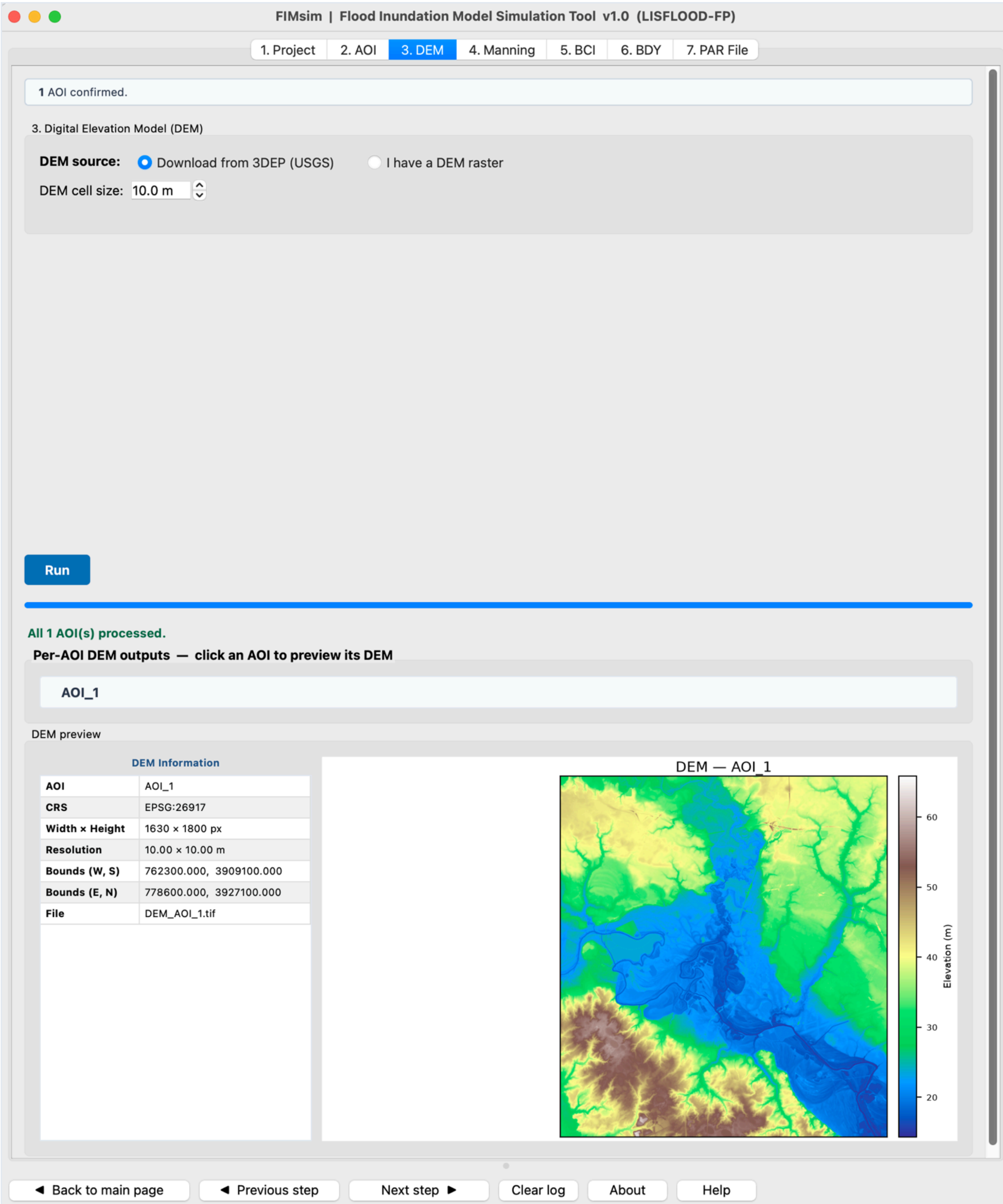
Expected: - Area **293.40 km²**, State **North Carolina (NC)**, CRS **EPSG:26917** - HUC6 **030202** | HUC8 03020201, 03020202, 03020203 - Main river: **Neuse River** - USGS gage found: **02089000 — Neuse River near Goldsboro, NC**



Step 3 — DEM

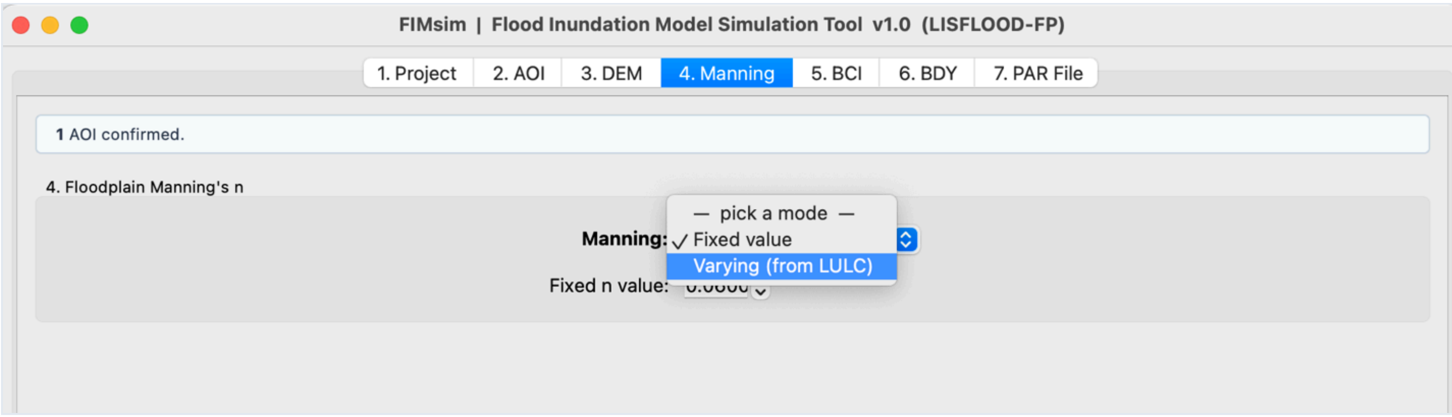
Keep **Download from 3DEP (USGS)** and set the DEM cell size to **10 m.** Click **Run.** FIMsim downloads the elevation tiles, mosaics, resamples, reprojects to the AOI's UTM zone, and clips to the AOI.

Expected: a 1630 × 1800 px DEM at 10 m resolution (`DEM_AOI_1.tif`), previewed with elevations of roughly 15–65 m, plus `dem.ascii` in `lisflood-files/`.

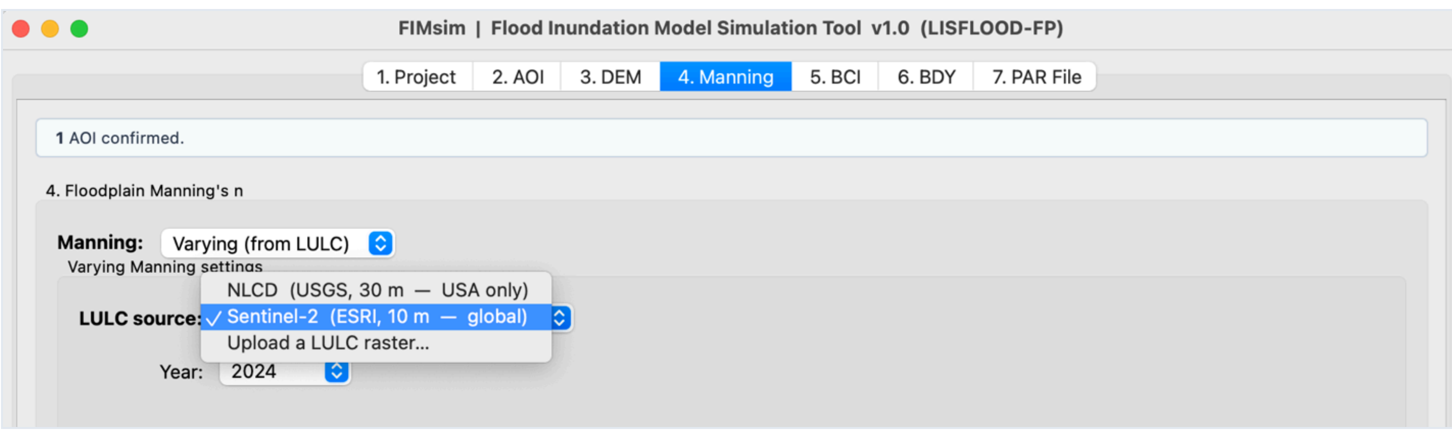


Step 4 — Manning

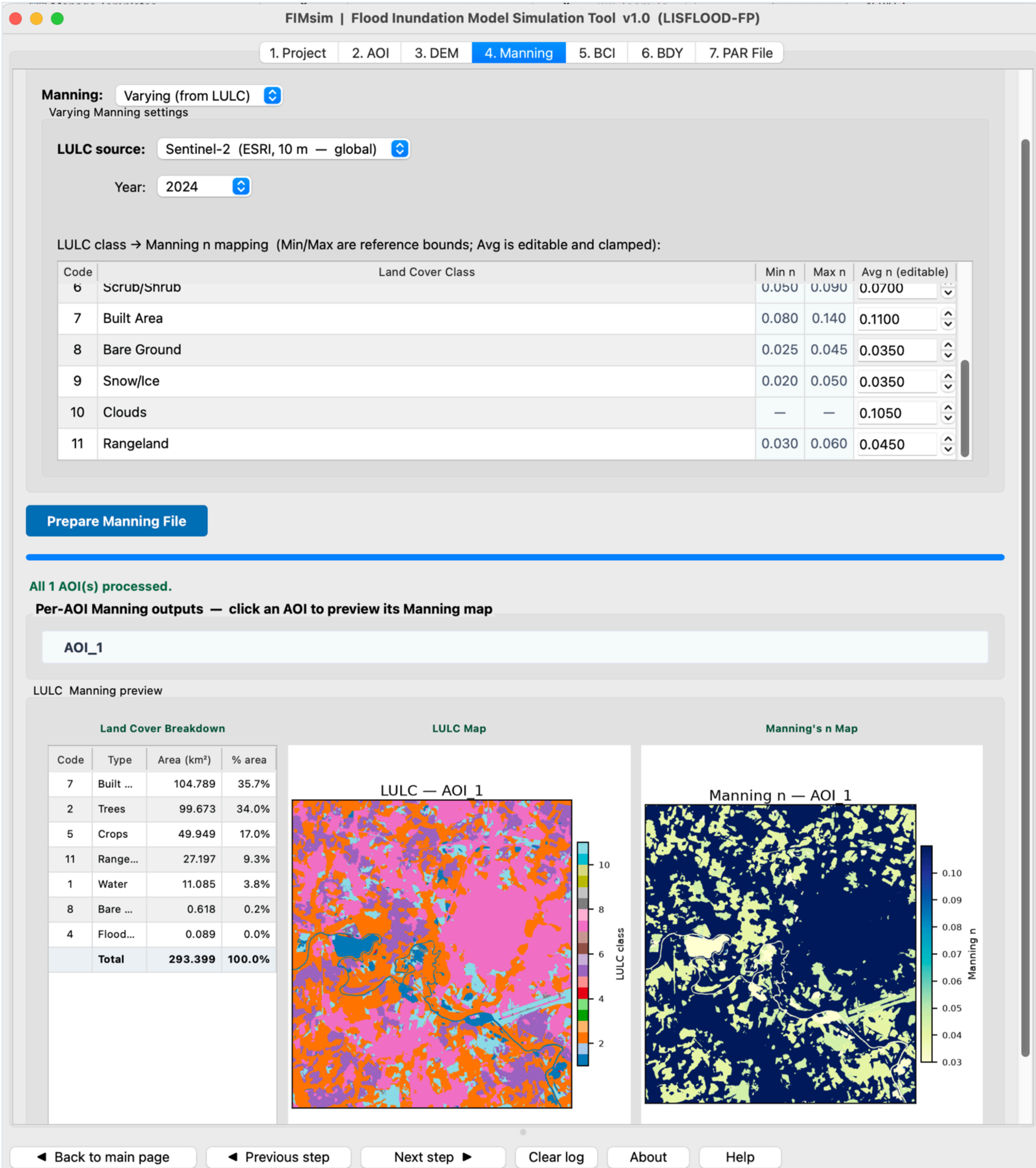
First choose between **Fixed** roughness (one value everywhere) and **Varying** roughness (derived from land cover):



Select **Varying**, then pick the LULC source and year — for this test case use **NLCD**. FIMsim downloads the land-cover raster, resamples/reprojects/clips it to the DEM grid, and converts it to a Manning's n map using the editable lookup table:



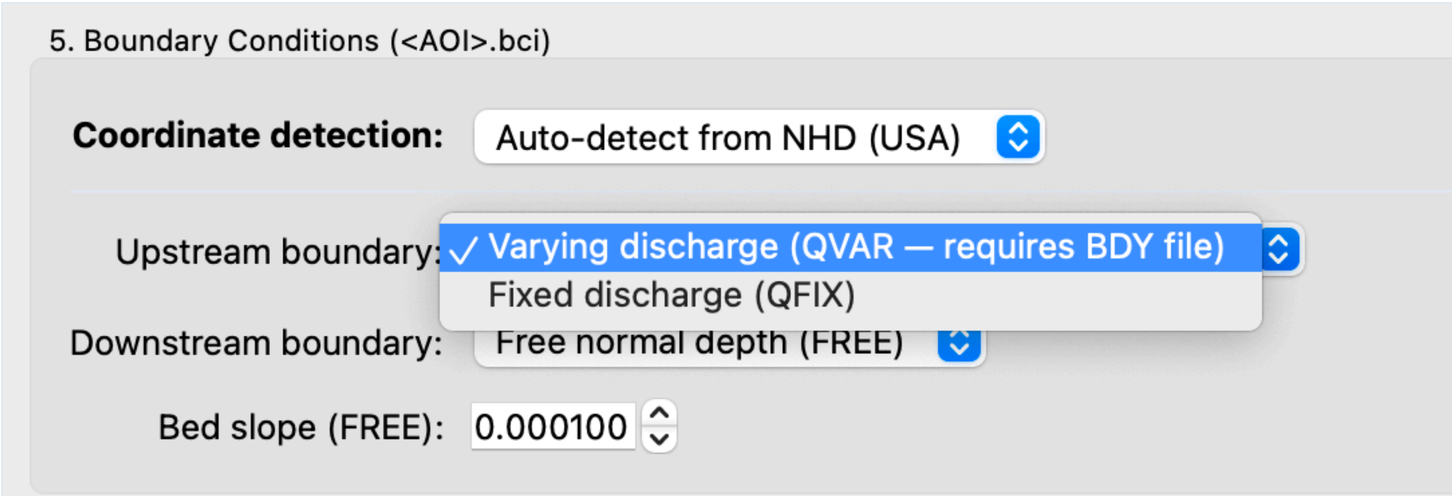
Expected: side-by-side LULC and Manning's n previews with per-class percentages (the AOI is dominated by cultivated crops and woody wetlands). Files produced: `LULC_AOI_1_<year>.tif`, `ManningN_AOI_1.tif`, and `lulc.ascii`.



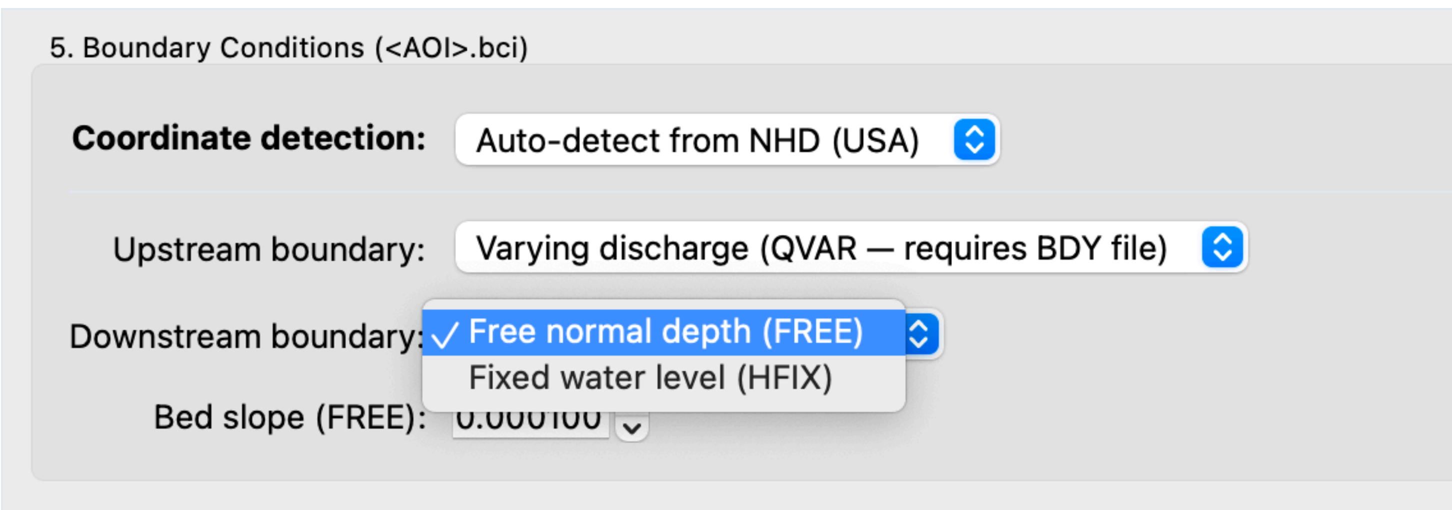
Step 5 — BCI (Boundary Conditions)

Keep **Auto-detect from NHD (USA)**.

The upstream boundary can be a **Varying discharge (QVAR)** — driven by the hydrograph from the next step — or a **Fixed discharge (QFIX)**. Select **QVAR**:

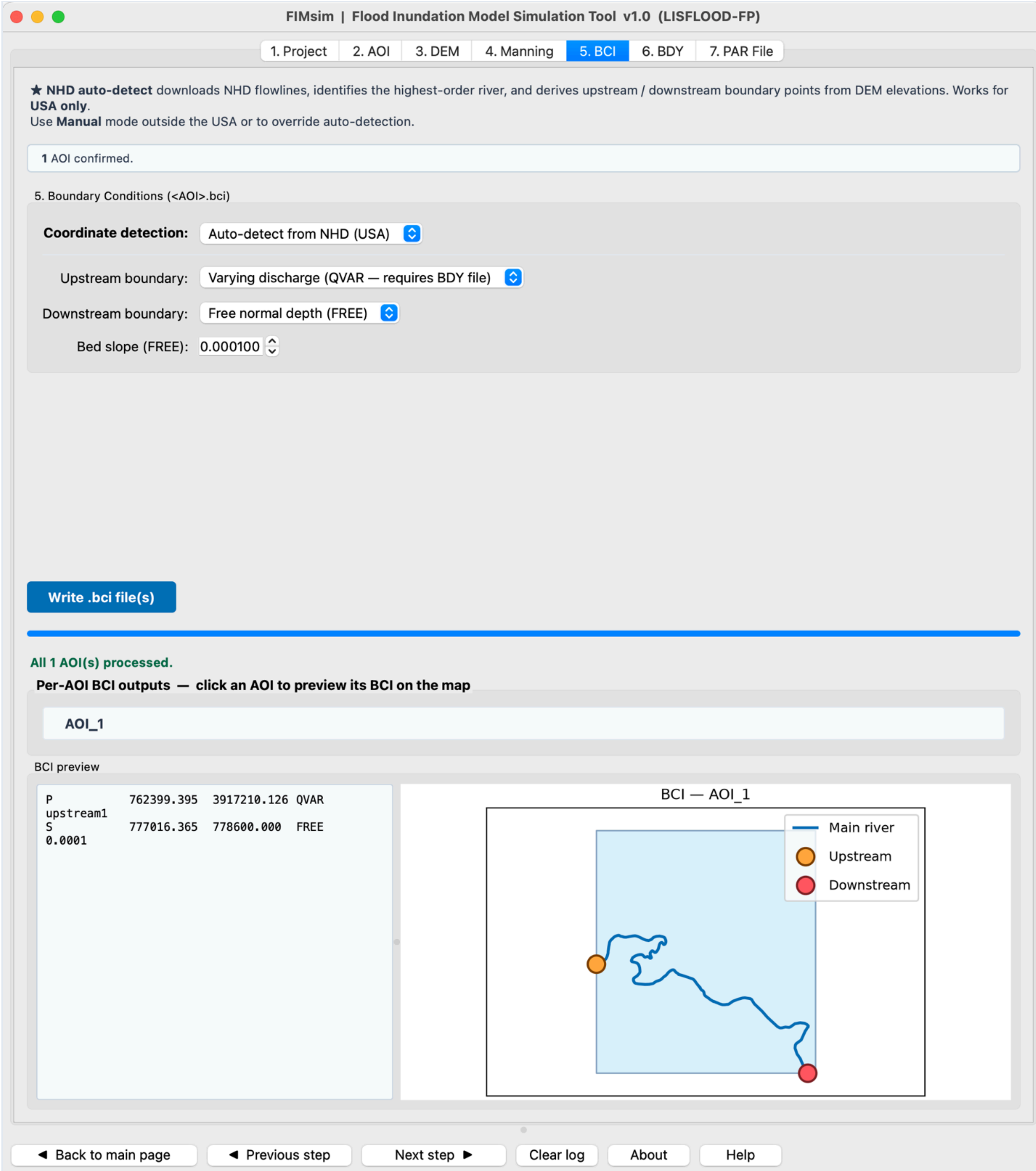


The downstream boundary can be a **Free normal depth (FREE)** or a **Fixed water level (HFIX)**. Select **FREE** with bed slope `0.0001`:



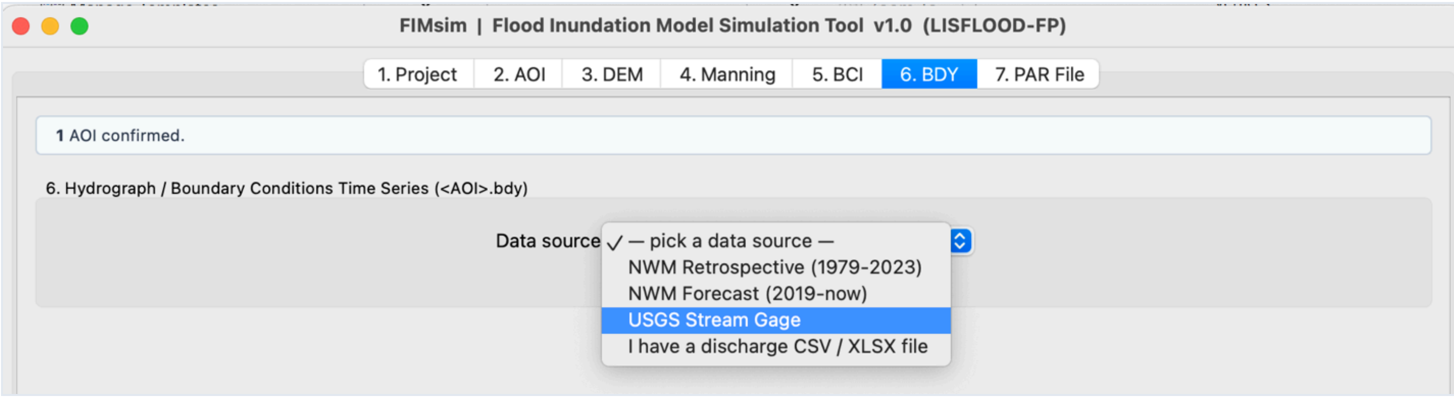
Click **Write .bci file(s)**. FIMsim downloads the NHD river network, identifies the main river, and derives the upstream/downstream boundary points from DEM elevations.

Expected: the BCI preview shows the Neuse River crossing the domain with the upstream point (orange) on the west edge and the downstream point (red) on the southeast corner, and the generated `AOI_1.bci` contains a `P … QVAR upstream1` line and an `S … FREE 0.0001` line.



Step 6 — BDY (Hydrograph)

The hydrograph step offers several data sources: **NWM Retrospective** (1979–2023), **NWM Forecast** (2019–present; short/medium/long range), **USGS Stream Gage**, or your own CSV file:

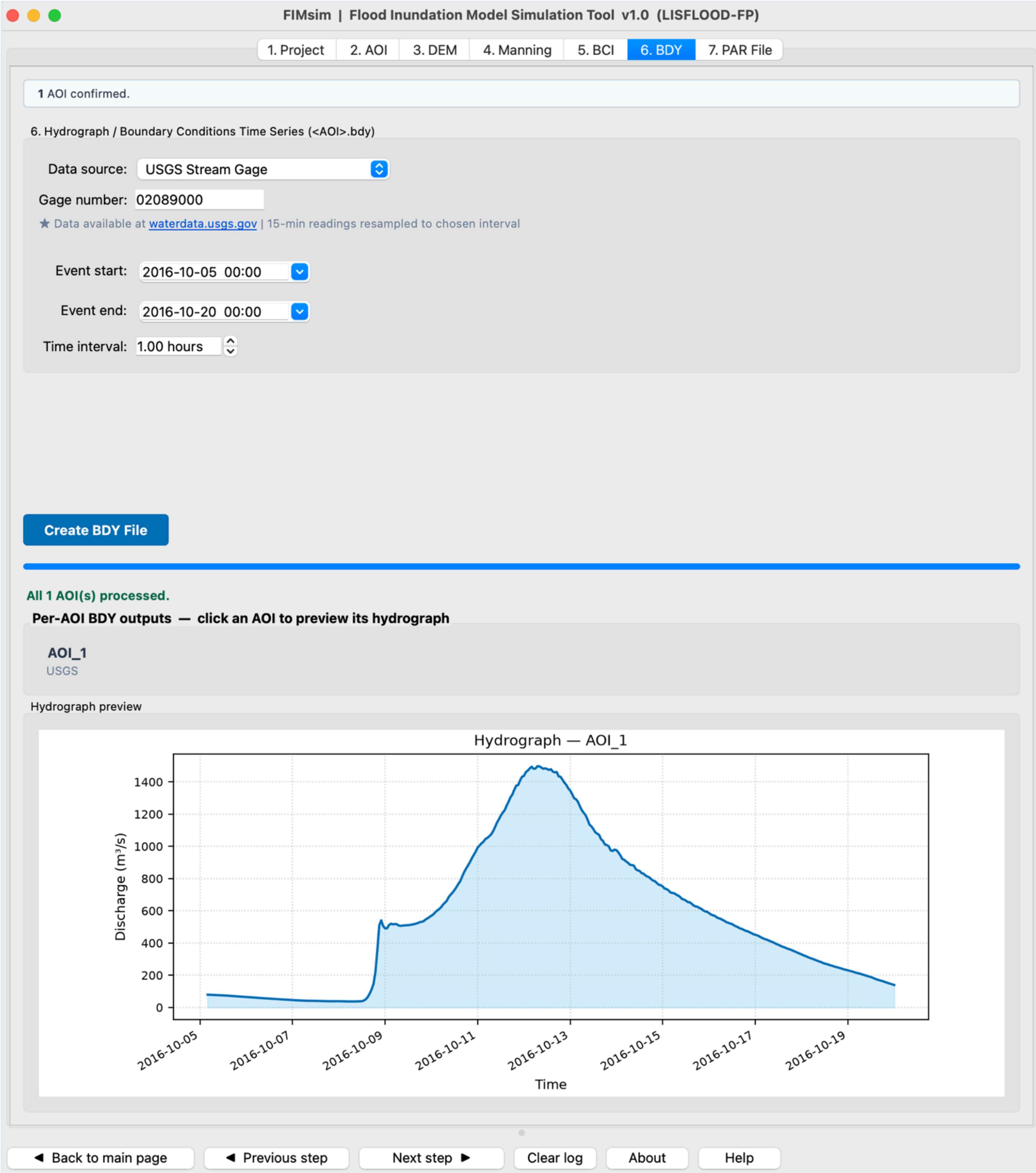


For this test case select **USGS Stream Gage** with the gage found in Step 2:

- **Gage number:** `02089000`
- **Event start:** `2016-10-05 00:00`
- **Event end:** `2016-10-20 00:00`
- **Time interval:** 1.00 hours

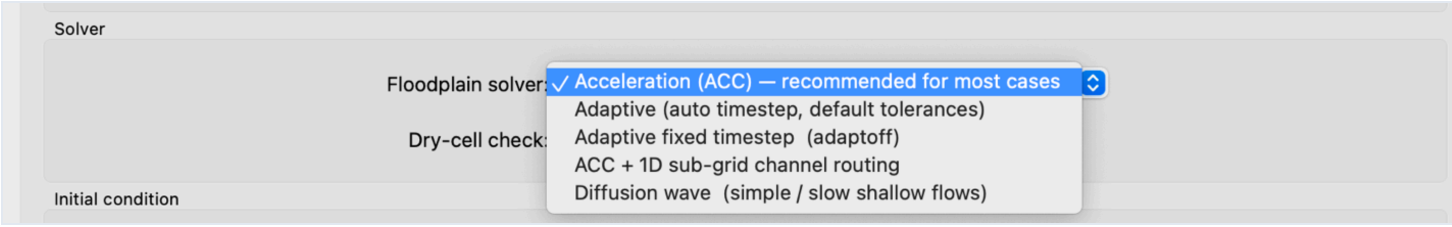
Click **Create BDY File**.

Expected: the hydrograph preview shows the Hurricane Matthew flood wave — rising sharply on **October 9** and peaking near **1,500 m³/s around October 12–13** — and `AOI_1.bdy` is written.

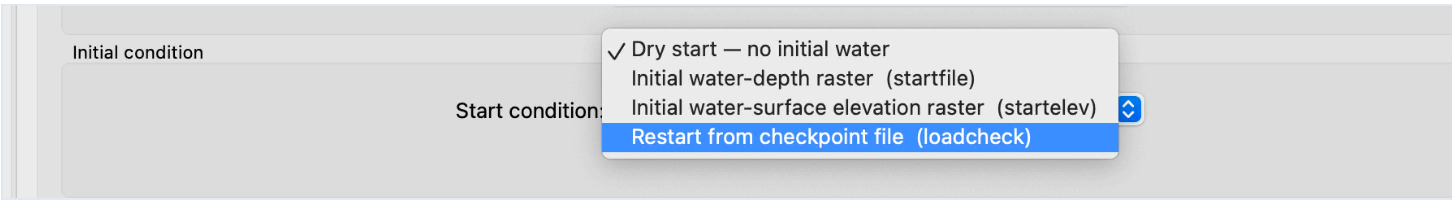


Step 7 — PAR (Parameter file)

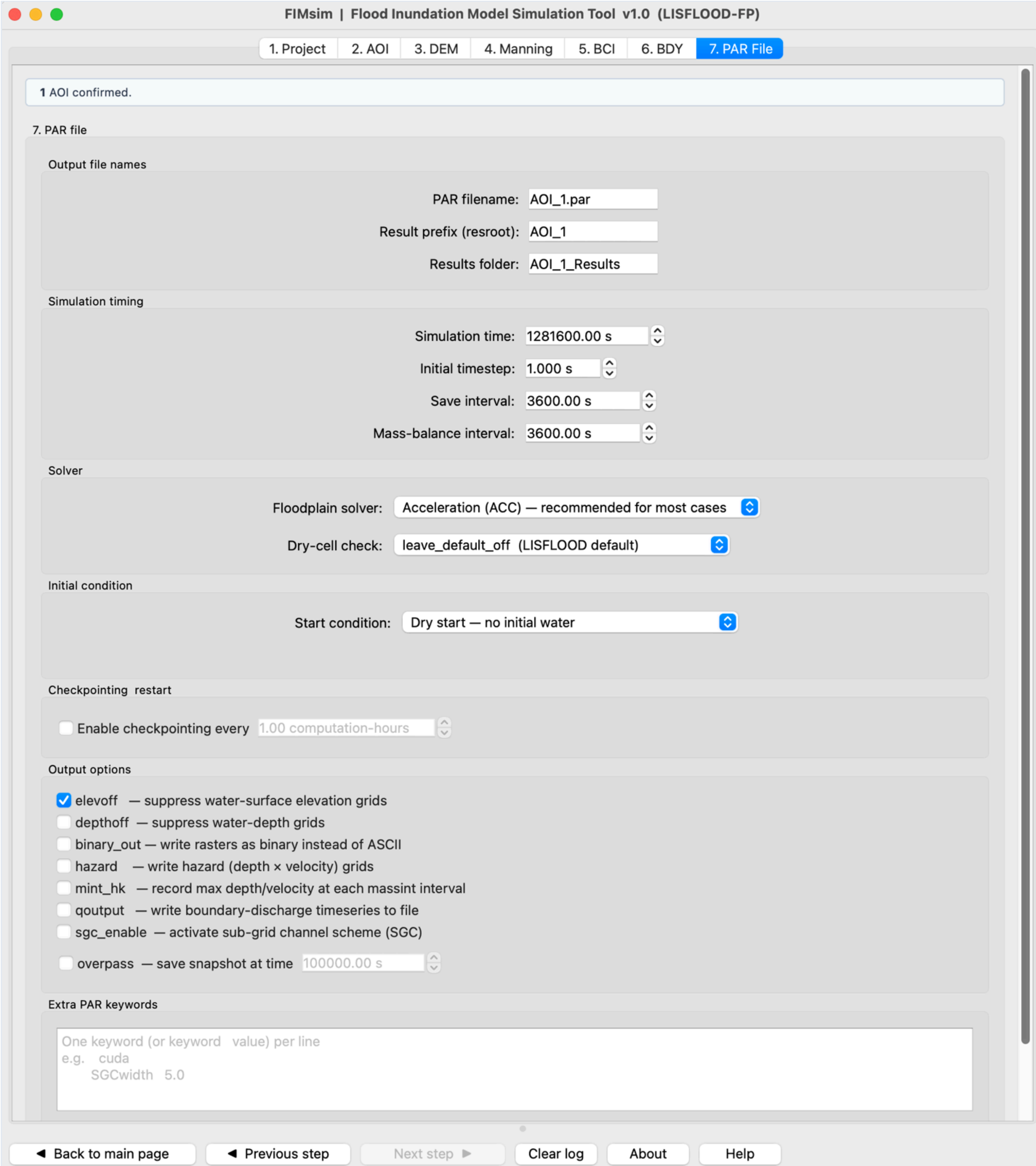
The final step assembles the LISFLOOD-FP parameter file. You can choose between solvers — the **Acceleration (ACC)** solver is recommended for most cases:



…and set the initial condition (keep **Dry start — no initial water**):



The simulation time is derived automatically from the hydrograph. Keep the remaining defaults (1 s initial timestep, 3600 s save interval) — there is also an *Extra PAR keywords* box for any additional LISFLOOD-FP keywords you may want:



Expected: `AOI_1.par` is written. The `lisflood-files/` folder now contains the complete LISFLOOD-FP input package — `dem.ascii`, `lulc.ascii`, `AOI_1.bci`, `AOI_1.bdy`, `AOI_1.par` — ready to run with any LISFLOOD-FP executable.

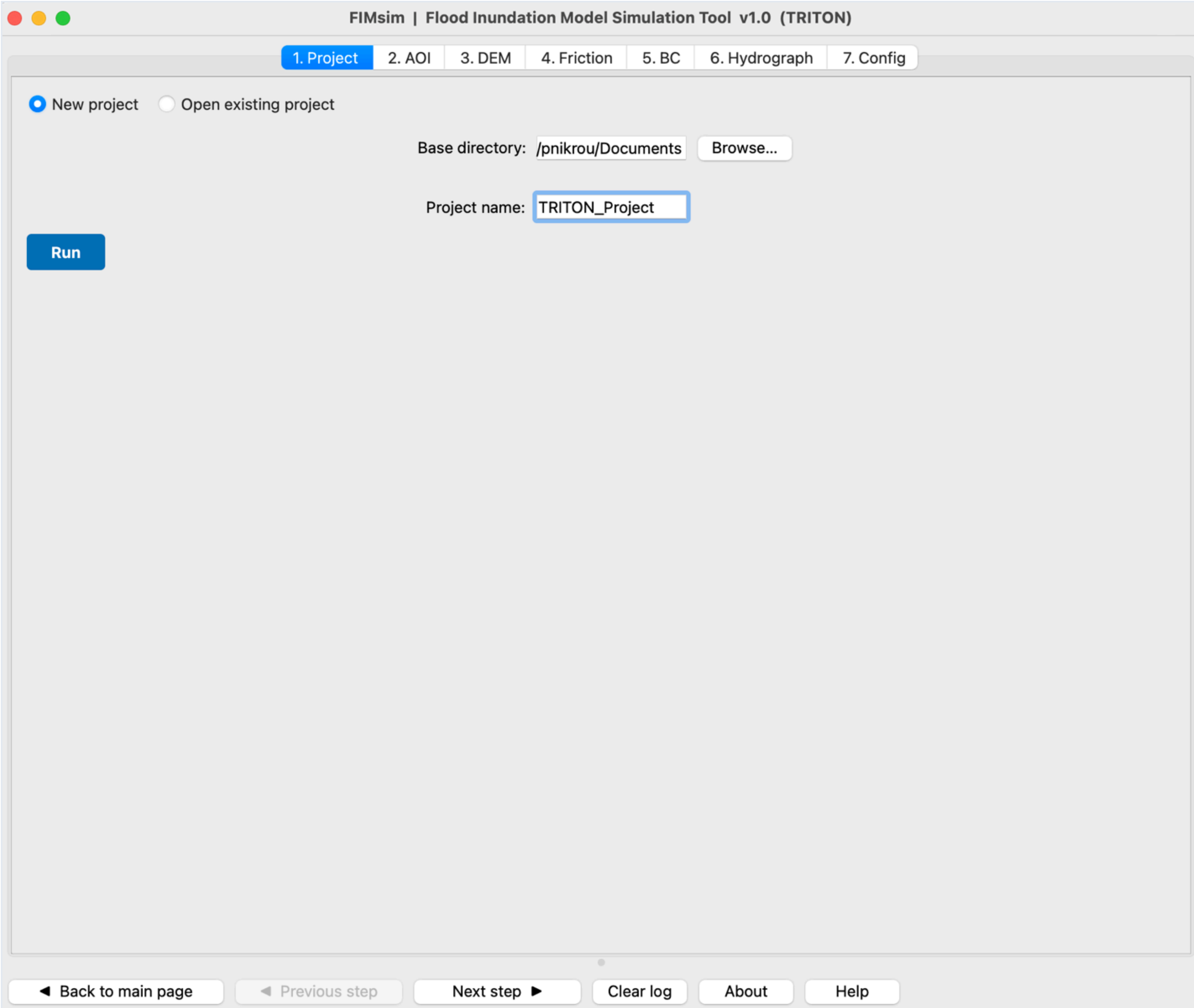
Test Case 2 — TRITON · Village Creek, Texas (Hurricane Harvey)

AOI: `AOI_2_Texas/AOI_2.shp` · 390.1 km² · CRS EPSG:26914 (NAD83 / UTM 14N) · Event: Hurricane Harvey, 2017-08-24 → 2017-09-10

In late August 2017, Hurricane Harvey stalled over southeast Texas and produced the heaviest tropical rainfall ever recorded in the United States. Village Creek — a tributary of the Neches River near Kountze, TX — experienced record flooding. This walkthrough builds the complete TRITON input package for that event using the NWM Retrospective discharge for the automatically detected reach.

Step 1 — Project

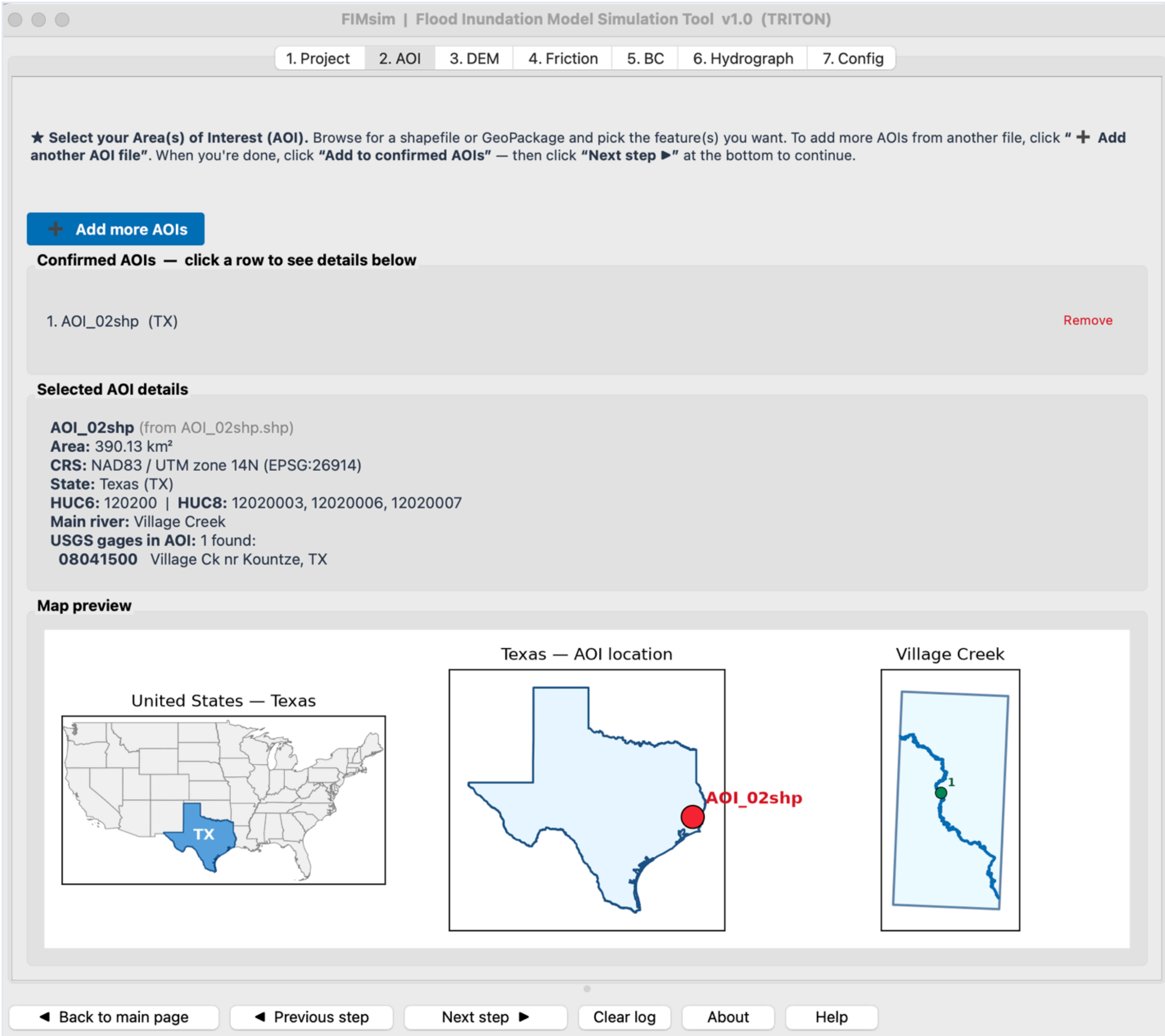
Open FIMsim and choose the **TRITON** model. Create a new project — pick any project name and an empty output folder.



Step 2 — AOI

Click **Browse** and select `AOI_2.shp`, tick the single feature, and click **Add to confirmed AOIs**.

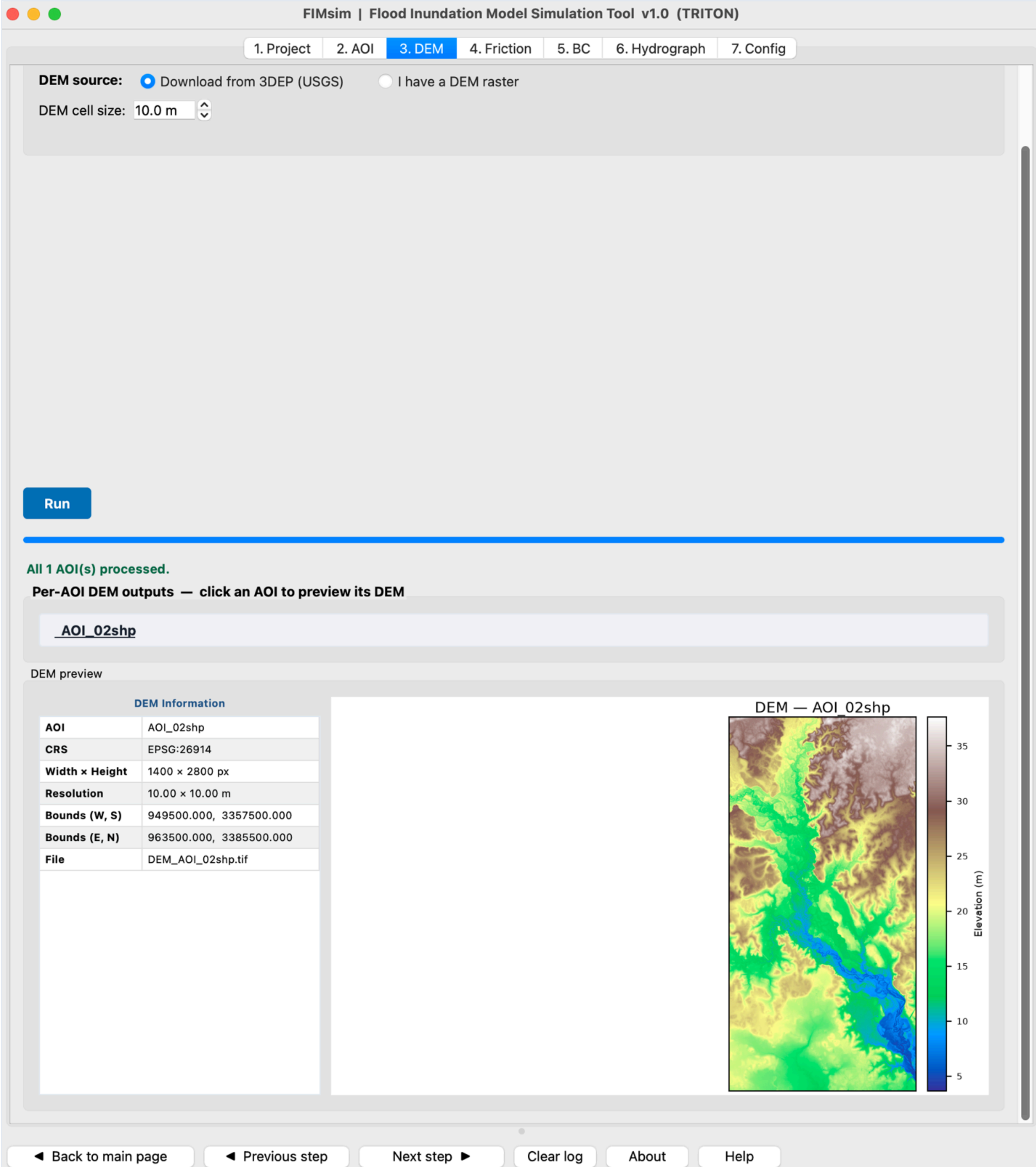
Expected: - Area **390.13 km²**, State **Texas (TX)**, CRS **EPSG:26914** - HUC6 **120200** | HUC8 12020003, 12020006, 12020007 - Main river: **Village Creek** - USGS gage found: **08041500 — Village Ck nr Kountze, TX**



Step 3 — DEM

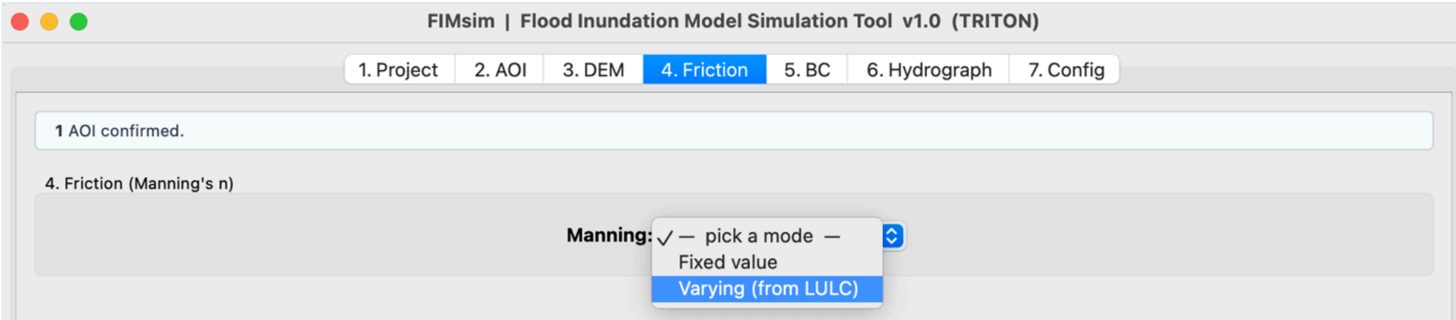
Keep **Download from 3DEP (USGS)** with a **10 m** cell size and click **Run**.

Expected: a 1400 × 2800 px DEM at 10 m resolution (`DEM_<AOI>.tif`), with elevations of roughly 3–38 m and Village Creek clearly visible as the low (blue) corridor, plus `dem.asc` in `triton-files/`.

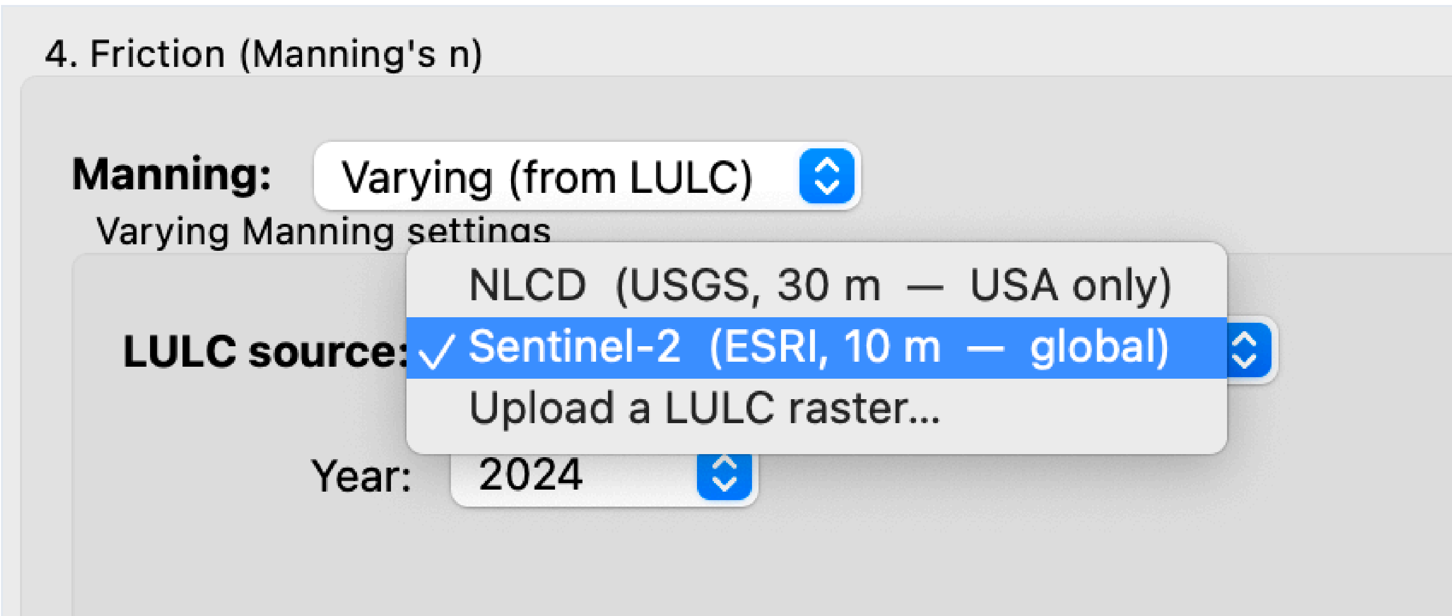


Step 4 — Friction

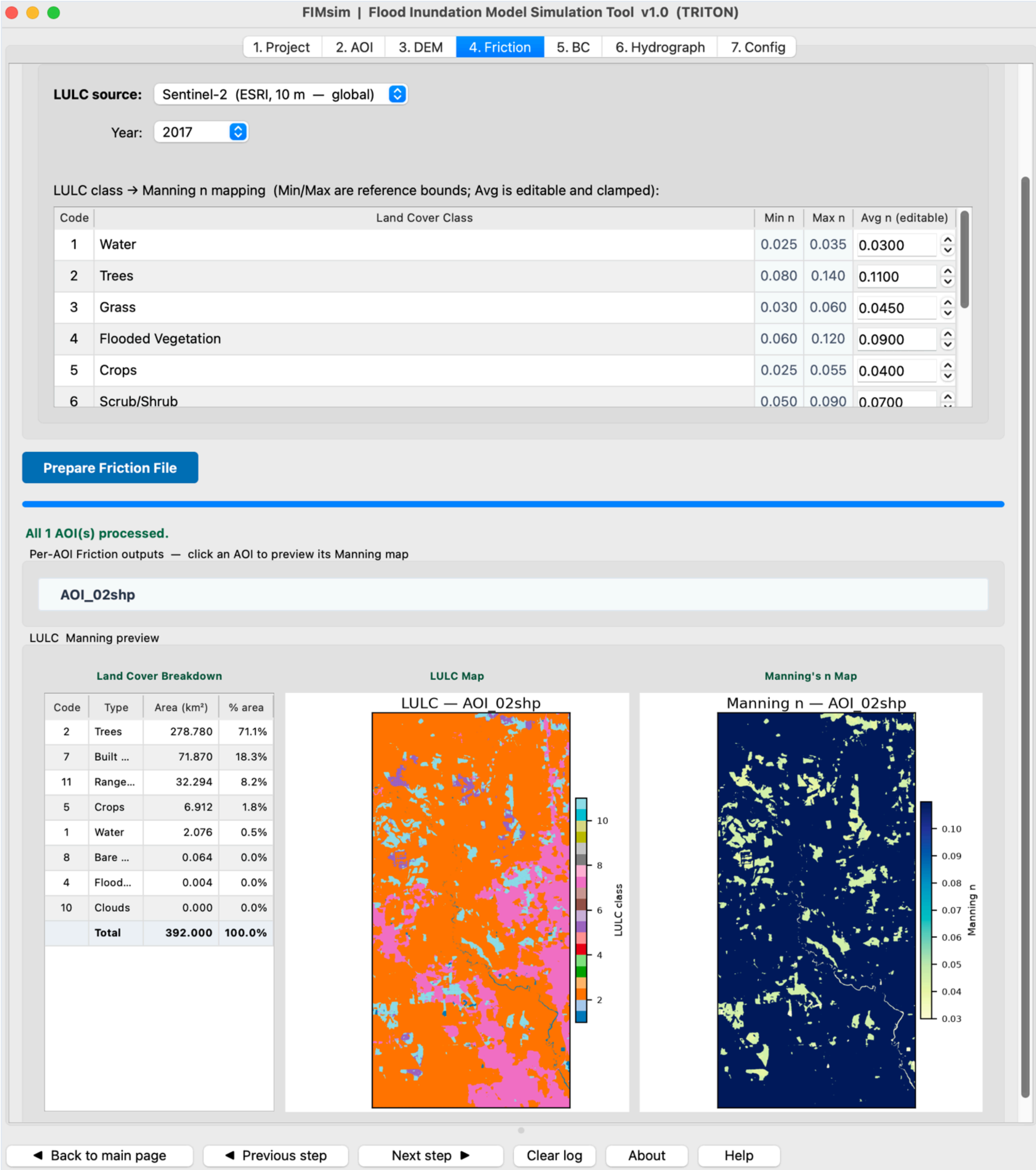
First choose between **Fixed** and **Varying** roughness:



Select **Varying**, then pick the LULC source and year (use **NLCD**):

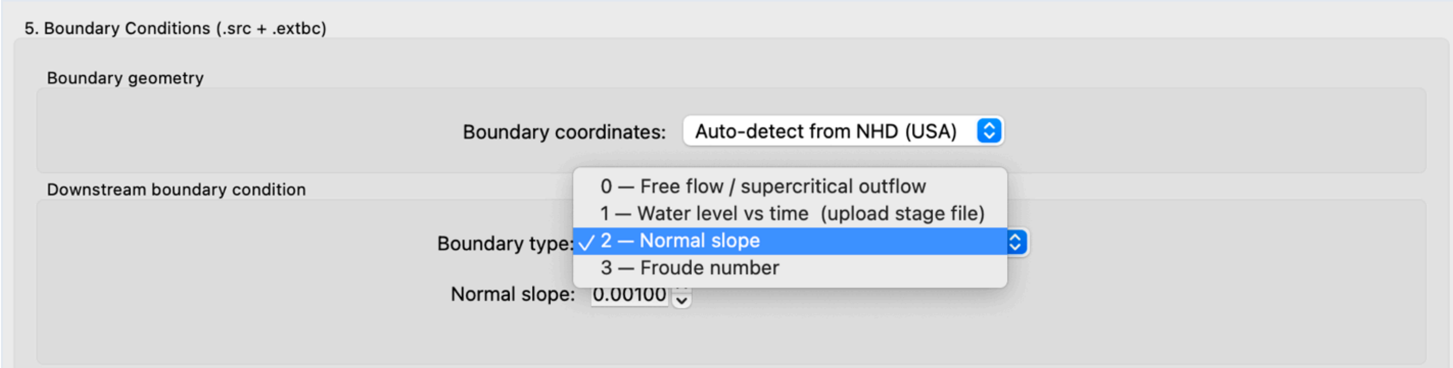


Expected: LULC and Manning's n previews (the AOI is dominated by forest and woody wetlands). Files produced: `LULC_<AOI>_<year>.tif`, `ManningN_<AOI>.tif`, and the TRITON friction raster `friction.asc` snapped to the DEM grid.



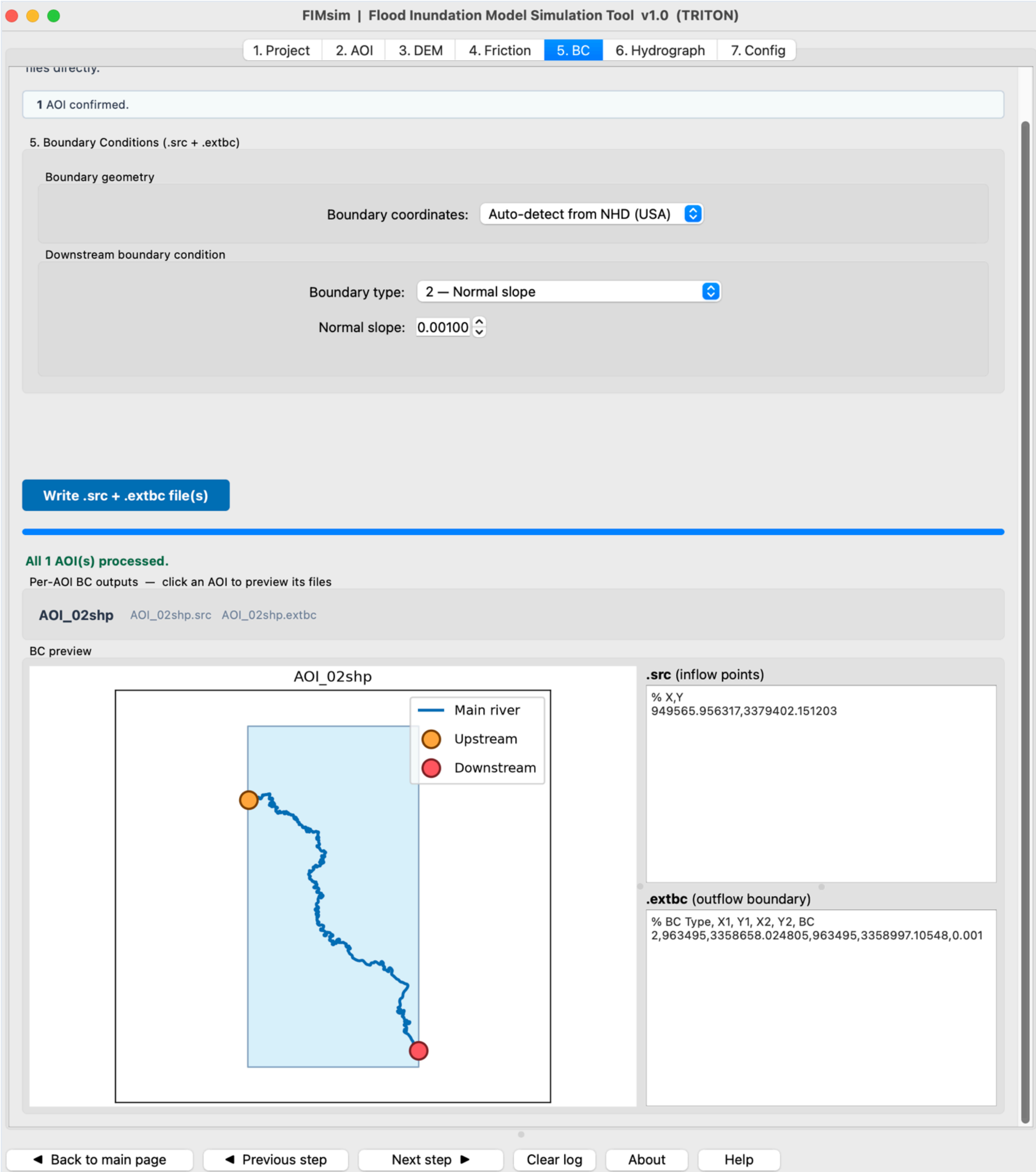
Step 5 — BC (Boundary Conditions)

Keep **Auto-detect from NHD (USA)**. The downstream boundary type defaults to **2 — Normal slope** with slope `0.001` — keep both:



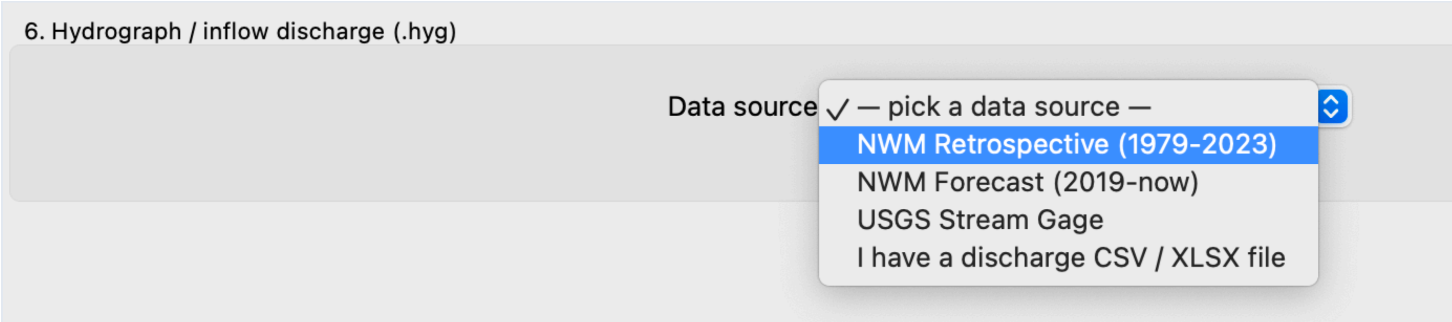
Click **Write .src + .extbc file(s)**.

Expected: the preview map shows Village Creek crossing the domain with the upstream inflow point (orange) on the northwest edge and the downstream boundary (red) at the southeast corner. Two files are generated and previewed: `<AOI>.src` (inflow point coordinates) and `<AOI>.extbc` (the type-2 outflow boundary segment on the DEM edge with slope 0.001).



Step 6 — Hydrograph

The hydrograph step offers the same data sources as LISFLOOD's BDY step — NWM Retrospective (1979–2023), NWM Forecast (2019–now), USGS Stream Gage, or your own CSV/XLSX file:

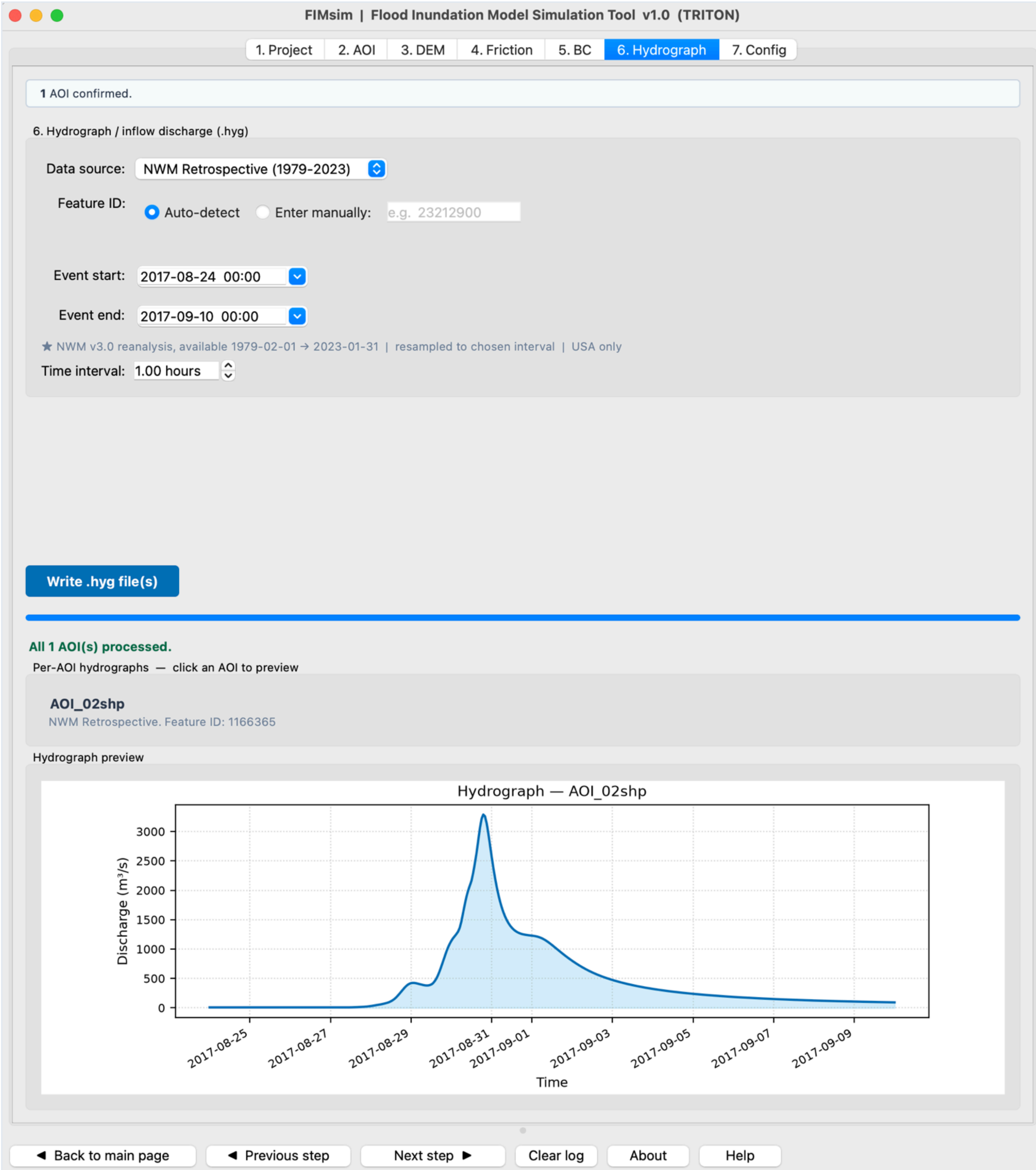


In Test Case 1 (LISFLOOD-FP) we used a **USGS stream gage**; this time we download the discharge from the **NWM** instead — simply to demonstrate both processes. Select **NWM Retrospective (1979–2023)** and keep **Feature ID: Auto-detect** (you can also enter a feature ID manually). Set:

- **Event start:** `2017-08-24 00:00`
- **Event end:** `2017-09-10 00:00`
- **Time interval:** 1.00 hours

Click **Write .hyg file(s)**.

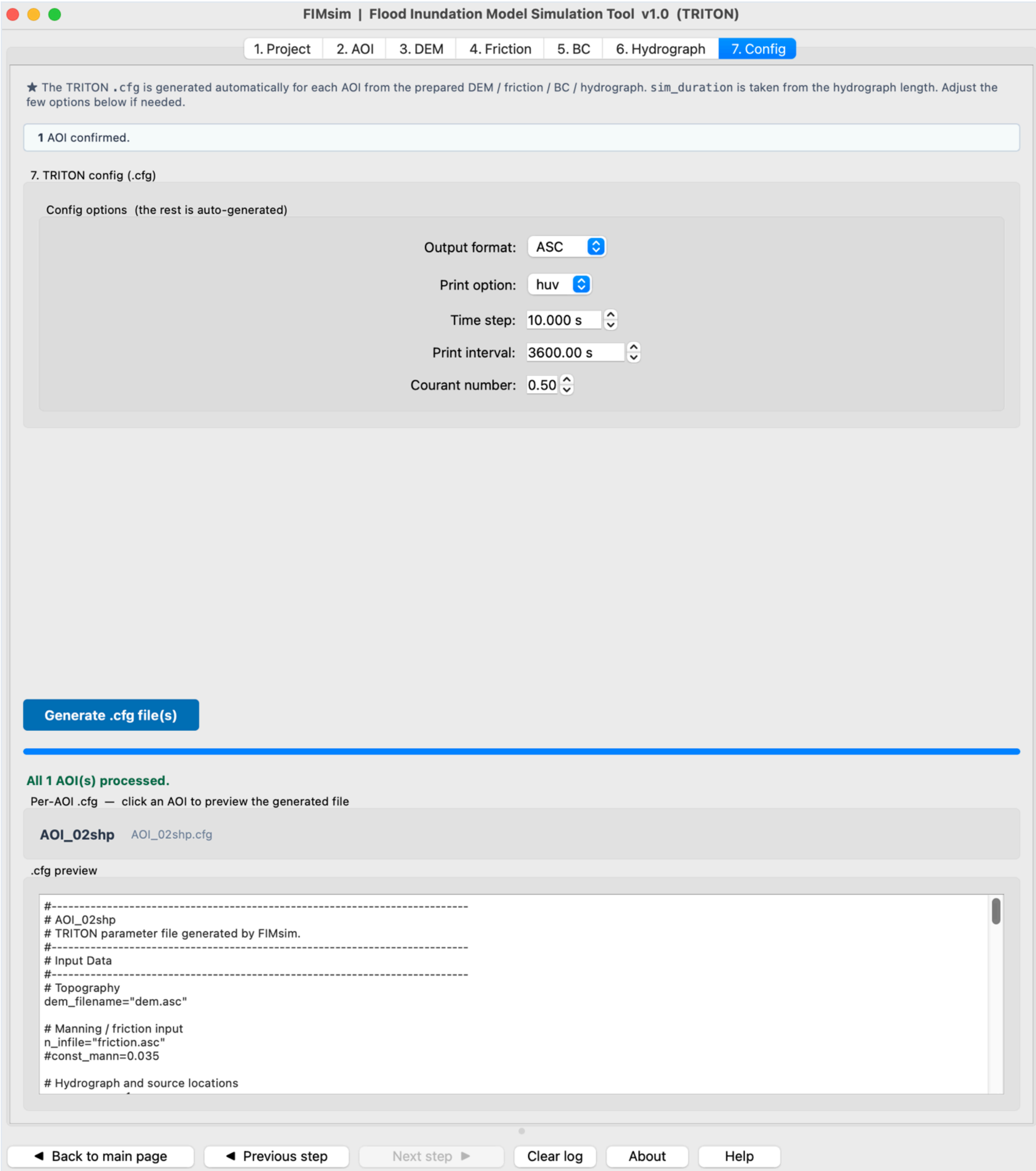
Expected: the auto-detected feature ID is **1166365** (shown under the AOI name), and the hydrograph preview shows the Hurricane Harvey flood wave peaking near **3,250 m³/s around August 31**. The `<AOI>.hyg` file is written.



Step 7 — Config

The final step assembles the TRITON configuration file. Keep the defaults (ASC output format, time step, print interval — the simulation duration comes from the hydrograph automatically) and run.

Expected: `<AOI>.cfg` is written. The `triton-files/` folder now contains the complete TRITON input package — `dem.asc`, `friction.asc`, `<AOI>.src`, `<AOI>.extbc`, `<AOI>.hyg`, `<AOI>.cfg` — ready to run with the TRITON solver.



Troubleshooting

- **A download step fails or times out** — the USGS / NOAA data services occasionally have outages; simply re-run the step.
- **NWM Retrospective date limits** — the archive covers **Feb 1979 – Jan 2023**; the NWM Forecast archive starts in 2019.
- **USGS gage data** — retrieved from waterdata.usgs.gov; 15-minute readings are resampled to your chosen time interval.